Week 12, Lecture 21
Complete Markets and Full Insurance

1 A two-period exchange economy

There are two consumers, A and B, and two dates, $t = 0, 1$. At date 1 one of two states, H or L, occurs, each with probability 1/2. There is one nonstorable consumption good. Both consumers have preferences

$$U^i = \ln c_0^i + \frac{1}{2} \ln c_1^i(H) + \frac{1}{2} \ln c_1^i(L), \qquad i \in \{A, B\}.$$

For simplicity, the subjective discount factor is one. Their endowments are

	Date 0	Date 1: H	Date 1: L
Consumer A	2	4	0
Consumer B	2	0	4
Aggregate	4	4	4

Individual endowments are risky, but aggregate endowment is constant. Without contingent trades, each consumer has zero consumption in one state; logarithmic utility makes that especially costly.

2 Arrow securities and the budget constraint

At date 0, the H-security pays one unit of consumption at date 1 if H occurs, and zero otherwise. The L-security pays one unit if L occurs, and zero otherwise. Their date-0 prices are q_H and q_L. Consumer i chooses positions a_H^i, a_L^i, where negative positions are sales:

$$\begin{aligned} c_0^i + q_H a_H^i + q_L a_L^i &= y_0^i, \\ c_1^i(H) &= y_1^i(H) + a_H^i, \\ c_1^i(L) &= y_1^i(L) + a_L^i. \end{aligned}$$

Combining the constraints yields

$$\boxed{c_0^i + q_H c_1^i(H) + q_L c_1^i(L) = y_0^i + q_H y_1^i(H) + q_L y_1^i(L).} \tag{1}$$

Complete markets means that there is a traded claim for consumption in every possible future state.

3 Consumer optimization

Write $W^i = y_0^i + q_H y_1^i(H) + q_L y_1^i(L)$. Consumer i maximizes utility subject to (1). The Lagrangian is

$$\mathcal{L}^i = \ln c_0^i + \frac{1}{2}\ln c_1^i(H) + \frac{1}{2}\ln c_1^i(L) + \lambda^i\left[W^i - c_0^i - q_H c_1^i(H) - q_L c_1^i(L)\right].$$

The first-order conditions are

$$\frac{1}{c_0^i} = \lambda^i, \qquad \frac{1}{2c_1^i(H)} = \lambda^i q_H, \qquad \frac{1}{2c_1^i(L)} = \lambda^i q_L.$$

Therefore,

$$\boxed{q_H = \frac{1}{2}\frac{c_0^i}{c_1^i(H)}, \qquad q_L = \frac{1}{2}\frac{c_0^i}{c_1^i(L)}.} \tag{2}$$

These are Arrow-security pricing equations: a state claim's price reflects its probability and the marginal utility of consumption in the state where it pays.

4 Competitive equilibrium

Goods-market clearing requires

$$c_0^A + c_0^B = 4, \qquad c_1^A(H) + c_1^B(H) = 4, \qquad c_1^A(L) + c_1^B(L) = 4.$$

The symmetric candidate allocation is

$$\boxed{c_0^A = c_0^B = 2, \qquad c_1^A(H) = c_1^A(L) = c_1^B(H) = c_1^B(L) = 2.} \tag{3}$$

By (2), the supporting prices are

$$\boxed{q_H = q_L = \frac{1}{2}.}$$

At these prices, each consumer's wealth is four:

$$W^A = 2 + \tfrac{1}{2}(4) + \tfrac{1}{2}(0) = 4, \qquad W^B = 2 + \tfrac{1}{2}(0) + \tfrac{1}{2}(4) = 4.$$

Each consumption bundle costs $2 + \frac{1}{2}(2) + \frac{1}{2}(2) = 4$. Thus individual optimization and goods-market clearing both hold.

5 The insurance portfolios

The required Arrow positions are consumption minus endowment in each state:

	H-security	L-security
Consumer A	-2	$+2$
Consumer B	$+2$	-2
Aggregate	0	0

Consumer A sells two H claims and buys two L claims. The net date-0 cost is $\frac{1}{2}(-2) + \frac{1}{2}(2) = 0$. Consumer B takes the opposite portfolio. In either state, each consumes two units:

$$c_1^A(H) = 4 - 2 = 2, \quad c_1^A(L) = 0 + 2 = 2,$$
$$c_1^B(H) = 0 + 2 = 2, \quad c_1^B(L) = 4 - 2 = 2.$$

The consumers have fully insured their idiosyncratic endowment risk.

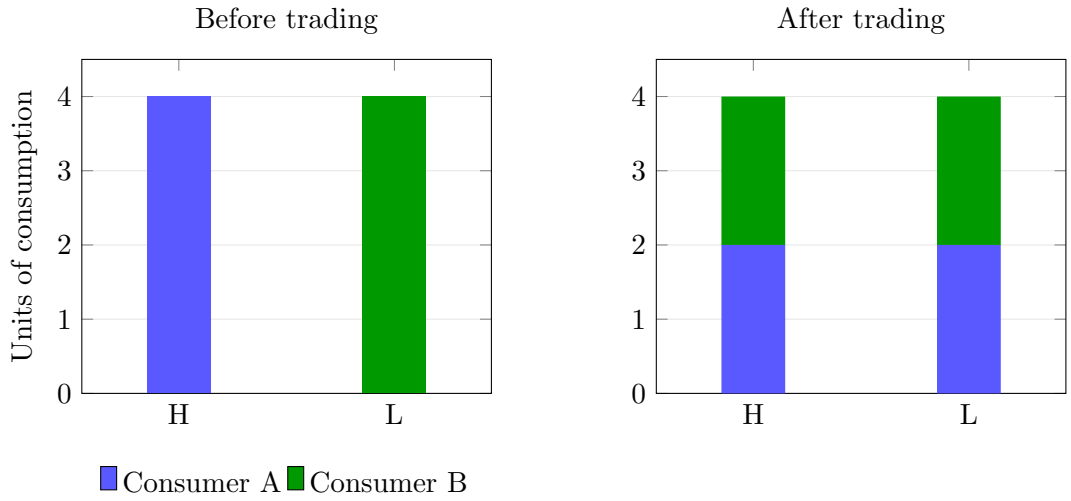


Figure 1: Complete markets eliminate individual endowment risk. Total consumption is four units in either state, while each consumer receives two units after trading.

6 Why insurance is optimal

Consider a Pareto planner with positive weights ω_A, ω_B. In state s, the planner chooses $c_1^A(s), c_1^B(s)$ to maximize

$$\omega_A \ln c_1^A(s) + \omega_B \ln c_1^B(s) \quad \text{subject to} \quad c_1^A(s) + c_1^B(s) = Y_1(s).$$

The first-order conditions give

$$\frac{\omega_A}{c_1^A(s)} = \frac{\omega_B}{c_1^B(s)} \quad \Longrightarrow \quad \boxed{\frac{c_1^A(s)}{c_1^B(s)} = \frac{\omega_A}{\omega_B}.}$$

Thus relative consumption is independent of the state. With equal weights, each gets $Y_1(s)/2$. If aggregate endowment is constant, individual consumption is constant too. If aggregate endowment varies, complete markets can share but cannot eliminate that aggregate risk.

7 Three lessons

1. Arrow securities make transfers contingent on the realized state.

2. State prices reflect probabilities and marginal utilities, and are pinned down jointly by optimization and market clearing.

3. Complete markets can fully insure idiosyncratic risk when aggregate resources are constant; aggregate risk remains when resources fluctuate.